

Robo - E : An Embodied Study Companion

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Abstract—ROBO-E is a multimodal educational robot designed to support students during independent learning. The system combines computer vision, natural-language interaction, and expressive robotic behaviors to create a more engaging study experience. An NVIDIA Jetson Nano serves as the primary computing platform for lightweight NLP and vision processing while an Arduino provides low-level servo and motor control. An Arducam enables visual perception, including user detection and tracking, while servos provide gentle head movements and blinking for nonverbal feedback. A touchscreen, microphone, and speaker provide additional interaction modalities. The system aims to investigate how embodied, responsive robotic interaction can support student engagement and learning.

I. INTRODUCTION

Code and files found here: https://github.com/raunika/ROB599_Design_HRI_Project_Umich

The increasing use of digital technologies in education has made learning resources more accessible and flexible for students. Online courses, educational applications, and tutoring allow students to access material outside of traditional classroom settings and, in some cases, provide personalized feedback based on their performance. However, effective learning is more than simply providing students with information. Interaction with an instructor and the perception that an instructor is present and responsive can play an important role in a student's learning process. Research on online learning has found positive relationships between social presence and student learning, suggesting that the social aspects of an educational environment remain important even when instruction is delivered digitally [1]. These findings motivate the exploration of educational technologies that can not only provide instructional content but also a more interactive and socially present learning experience. Social robots are particularly interesting in this context because they can combine the adaptability of digital systems with a physical embodiment capable of participating in face-to-face-like interactions. Unlike a conventional computer interface, a robot can occupy the learner's physical environment and communicate through multiple modalities, including speech, gaze, gestures, and other social behaviors. Previous research has also demonstrated the potential of social robots as tutors and has explored how different tutoring and interaction strategies can influence the student's learning experiences [2], [3]. These studies suggest that the behavior of the robot and the way students interact with it are important

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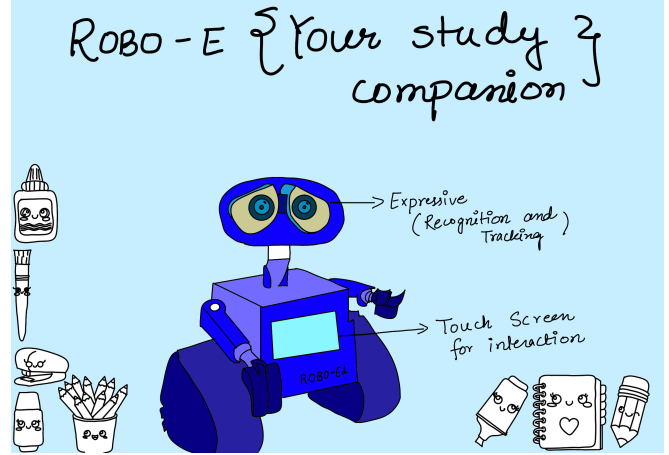


Fig. 1. Robo E - low fidelity desktop study companion

considerations when designing educational robots. Building on this previous work, our project explores a social robot that acts as an interactive learning companion for teaching a new concept to a student. The robot will be designed to explain concepts, ask questions, assess the student's understanding, and provide appropriate guidance or hints when the student encounters difficulty.

The goal of this project is to investigate how students experience learning through an embodied robotic tutor. In particular, we are interested in whether the robot's interactive behavior can create a learning experience that students perceive as helpful and natural.

Therefore, this project investigates the following research question:

Research Question: How do students perceive the experience of learning a new concept with a social robot tutor?

II. RELATED WORK

Previous research has explored how social robots can support tutoring and how the design of their interactions influences student's learning. Ramachandran et al. investigated how a robot tutor could shape productive help-seeking behavior during tutoring interactions with children [2]. Rather than simply providing help whenever it was requested, their robot used adaptive strategies to encourage a more productive way of seeking help. Their findings suggest that a robot can play an active role in shaping how students approach the learning process as a whole rather than seeing it as simply delivering information. Jordan et al. further examined how the method through which students requested

assistance influenced their interaction with a robot tutor [3]. In this study participants were provided with different ways to request assistance, including an on-screen button, a physical button, and raising a hand. The study found that these different help-seeking methods influenced participant's perceptions of the usefulness and benefits of the robot tutor. This highlights the importance of considering the interaction mechanism itself when designing educational robots. More recent work by Hei et al. compared individual tutoring with a human tutor and with a social robot, examining the participant's performance and cognitive load [4]. Their results showed that participants sought more help from the robot than from the human tutor. However, participants also experienced greater cognitive load when interacting with the robot. These findings demonstrate the potential and challenges of using social robots for education. This study shows us that encouraging interaction alone is not sufficient and that the design of the tutoring experience as a whole must also support the student.

III. TECHNICAL SUMMARY

A. Robot Concept

Robo-E is an interactive education robot designed to support students during their independent study sessions. We plan to use a camera based vision system that enables recognition and tracking of the students. We also hope to incorporate a touch screen display for an interactive access to the learning content, multiple choice questions and flashcards for active recall. Additionally, Robo-E uses Integrated speech input/output for natural language communication between student and robot educator.

Robo - E uses Jetson Nano as the main onboard computer. It is used to run lightweight AI/NLP, and vision processing. Arducam camera sensor is used for facial detection, user presence detection, user tracking etc using OpenCV and other Python vision libraries. SG90 servos will be used for gentle head orientation, blinking and other eye mechanisms and to provide other non-verbal expressive feedback. Arduino microcontroller will be used for real - time low level servo control.

A microphone + speaker module will support speech input/output with jetson handling the high level processing. Robo E is a desktop companion, hence we will be using wired power source.

B. Work Plan Timeline

We plan to divide the project into three main phases with each team member working on one or two of the sub-parts. These sub-parts are the physical design and embodiment of the robot, the chatbot driving the interaction, the perception enabling the robot to dynamically interact with the user, and the quantitative analysis iteratively allowing us to make informed embodiment design choices. In phase one we intend to prototype the basic embodiment or major parts of it, such that we can gauge our user group's feeling around its effectiveness. With this data we can create primitive prototypes of the chatbot and perception.

Phase two - we aim to intelligently build up upon our initial prototype, incorporating user feedback into an under-actuated embodiment. The chatbot will start to be customized to the nuance of our problem, and the perception will start to be integrated into the physical robot instead of just simulated. With this updated prototype, we will go back to our users and evaluate the effectiveness of these changes and increased actuation.

Finally for phase three, we plan to tie everything together with a finalized embodiment, fully integrated perception, and a fully functional chatbot. With this we will be able to gather a final user evaluation.

In terms of timeline, we intend to finish phase one and as much of phase two as is possible before the prototype show off on October 27th. After receiving advice from that event, we will then aim to finish phase two within a week afterwards, before or on November 3rd. With phases one and two completed, phase three and our presentation slides will need to be completed by November 16th for the project presentation and live demo. Logistically, we will rotate who is working on each specific sub-part each phase, and each team member will work together on overarching decisions and ideation.

C. Anticipated Challenges

One anticipated challenge is running an AI model locally and making it work efficiently for our needs could be a challenge. For this project, we hope to make our robot expressive and use an underactuated system to accomplish this goal; use of gears and bar mechanisms can cause a delay in response time.

Additionally, the queue system used for all university 3D printing services. If each part takes several days to be printed, this could slow down the project substantially. Keeping this in mind, we will be diligent to submit printing requests ahead of time and to use personal 3D printing resources if needed. Furthermore, it does take time to both coordinate and schedule user studies; which need to be done in a timely manner such that we can stay on track and not have to delay embodiment development due to a lack of user feedback.

Another challenge could be the pedagogical method used to teach the users a concept, since the method used for one concept may not be the best way to teach another concept. To address this variable in this study, we have decided to teach a single concept and hence use a single method of teaching which we deem would be the most relevant.

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